to adapt dynamically to complex environments. Computer vision technologies support precision navigation and autonomous landing, while swarm intelligence principles enable coordinated operations among multiple drones for surveillance, mapping, or logistics. Furthermore, AI integration with cloud computing and edge processing expands analytical capabilities, allowing for real-time data interpretation and predictive maintenance.

1.4 Generation breakdown

The evolution of drone technology can be categorized into five distinct generations, each marking a significant advancement in design, control, and operational capability. These generations reflect the transition from simple remotely piloted aircraft to fully autonomous aerial systems integrated with artificial intelligence and networked operations.

1.4.1 First generation: Manual drones

The first generation of drone technology marks the formative stage in the evolution of unmanned aerial systems (UAS), characterized by rudimentary design, limited functionality, and entirely manual control. These early drones, developed primarily during the early to mid-20th century, served as experimental platforms to test the concept of flight without an onboard pilot. Their primary use was in the military domain, especially for target practice and reconnaissance during wartime, where they acted as substitutes for manned aircraft in hazardous or repetitive missions [Simic Milas, et al. 2018].

Technologically, first-generation drones were manually operated through basic radio frequency control systems that allowed limited maneuverability within a short range. They lacked the sophisticated avionics, sensors, and navigation systems seen in later generations. Flight stability and precision